# Deep Learning Approaches for Breast Cancer Detection, Segmentation and Classification: A Comprehensive Study

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**Abstract**

Breast cancer remains one of the most prevalent and deadly cancers affecting women worldwide. Early detection is crucial for improving survival rates and treatment outcomes. This study comprehensively evaluates various deep learning approaches for breast cancer detection, classification, and segmentation tasks. We analyze over 55 research papers and multiple GitHub repositories, identifying key challenges in medical image analysis for breast cancer. The experimental results demonstrate that advanced architectures like MedSAM with ViT backbones and attention-based U-Net variants achieve superior performance. The study highlights the critical need for accessible code and reproducible research in medical AI applications.

## 1 Introduction

Breast cancer represents a significant global health challenge, being the most commonly diagnosed cancer among women and a leading cause of cancer-related mortality worldwide. According to global cancer statistics, breast cancer accounts for approximately 25% of all cancer cases and 15% of cancer deaths in women. The five-year survival rate for breast cancer detected at an early localized stage exceeds 90%, compared to about 27% for cancers detected at advanced stages, underscoring the critical importance of early detection.Traditional breast cancer diagnosis relies on mammography, ultrasound, magnetic resonance imaging (MRI), and histopathological analysis. The Breast Imaging Reporting and Data System (BI-RADS) provides a standardized classification system ranging from 0 (incomplete) to 6 (known biopsy-proven malignancy), with category 4 representing "suspicious" findings that require further investigation. This category is particularly challenging due to its inherent ambiguity and variability in interpretation. Recent advances in deep learning have shown remarkable potential in automating and improving the accuracy of breast cancer detection and classification. This project provides a comprehensive analysis of state-of-the-art deep learning approaches for breast cancer analysis, evaluating both published research and practical implementations.

# 2 Project Description and Tasks

There are mainly three computer vision tasks in breast cancer analysis:

## 2.1 Classification

Determining the presence and type of breast cancer, including: Binary classification (benign vs. malignant), Multi-class classification (BI-RADS categories) and Histopathological image classification

## 2.2 Segmentation

Identifying and delineating regions of interest: Mass segmentation in mammograms, Lesion segmentation in ultrasound images and Whole slide image segmentation for biopsy analysis

## 2.3 Object Detection

Localizing suspicious regions: Mass detection and localization, Microcalcification detection and Lesion boundary detection

# 3 Challenges in Breast Cancer Analysis via Deep Learning

Data Scarcity and Accessibility Medical imaging datasets for breast cancer are often limited due to:

- Privacy concerns and regulations
- Cost and expertise required for annotation
- Restricted access to clinical data
- Institutional barriers to data sharing

Image Complexity Breast images present unique challenges:

- High resolution requirements (whole slide images)
- Variability in imaging modalities (mammography, ultrasound, MRI)
- Subtle and diverse appearance of abnormalities
- Class imbalance (few positive cases)

Clinical Validation

- Need for robust performance across diverse populations
- Requirement for high sensitivity in clinical settings
- Integration with existing clinical workflows
- Regulatory approval processes

# 4 Related Work

The field of deep learning for breast cancer analysis has witnessed remarkable growth, with numerous publications reporting exceptional performance metrics.However, after a comprehensive survey of over 55 recent papers revealed a critical reproducibility crisis: only two publications provided accessible code implementations, both demonstrating limited performance (62.5% accuracy on one of them and IoU of 44% for both), while the remaining papers despite reporting accuracies frequently exceeding 95% offered no publicly available code. This substantial gap between published results and reproducible implementations severely impedes scientific progress and independent validation of claimed advancements.

Current approaches in the literature predominantly feature convolutional neural networks (CNNs) for classification tasks, U-Net architectures for segmentation, and YOLO/Faster R-CNN frameworks for object detection. More recently, vision transformers (ViTs) have emerged as promising alternatives for various breast cancer analysis applications. Ensemble methods and hybrid architectures combining multiple approaches have also demonstrated competitive performance, though the inability to independently verify these results remains a significant concern for the research community.

# 5 Datasets

There are several publicly available breast imaging datasets:

**INbreast** [6]: A full-field digital mammography dataset containing 115 cases with 410 images, including mass annotations and BI-RADS classifications.

**CBIS-DDSM** [3, 4]: The Curated Breast Imaging Subset of DDSM contains about 10,000 images with mass and calcification annotations.

**BUSI** [1]: Breast Ultrasound Images Dataset containing 780 images with categories normal, benign, and malignant masses.

**VinDr-Mammo** [7, 8]: A large-scale dataset with 20,000 mammography examinations annotated for mass detection and BI-RADS assessment.

**MIAS** [9]: The Mammographic Image Analysis Society database containing 322 digitized mammograms with radiologist annotations.

# 6 Evaluation Metrics

Standard evaluation metrics:

- **Accuracy**: Proportion of correct predictions
- **AUC-ROC**: Area under the Receiver Operating Characteristic curve
- **Precision and Recall**: Trade-off between false positives and false negatives
- **F1-Score**: Harmonic mean of precision and recall
- **IoU (Intersection over Union)**: Area of overlap divided by area of union

- **Dice Coefficient**: Similarity measure between predicted and ground truth masks
- **mAP (mean Average Precision)**: Average precision across multiple IoU thresholds
- **Precision-Recall curves**: Performance across confidence thresholds

# 7 Results

## 7.1 Literature Survey Results

my comprehensive survey of 55+ publications (detailed in Appendix) revealed that while many papers report high accuracy (often exceeding 95%), very few provide accessible code for reproduction. The highest reported accuracies reached 99.12% for classification tasks, though these results should be interpreted with caution due to potential dataset biases and evaluation methodologies.

## 7.2 GitHub Repository Evaluation

Due to the fact that there is no available good code, an evaluation on famous GitHub repository is needed.

Table 1: Comparison of breast cancer detection models and their performance

| **Model & Repo** | **Dataset(s)** | **Task** | **Key Metrics** |
|---|---|---|---|
| YOLOv8-Medical-Imaging [2] (25M) | BUSI | Segmentation | Dice=0.76, IoU=0.613, F1=0.76 |
| breast_cancer_detection [5] (Detectron2, 41M) | InBreast, CBIS-DDSM, MIAS | Object Detection | Recall=0.33 |
| breast_cancer_detection [5] (ResNet-18, 11M) | InBreast, CBIS-DDSM, MIAS | Classification | Accuracy=0.68, F1=0.65 |
| breast_cancer_detection [5] (YOLOS, 11M) | InBreast, CBIS-DDSM, MIAS | Object Detection | Recall=0.36 |
| breast_cancer_detection [5] (YOLOL, 43M) | InBreast, CBIS-DDSM, MIAS | Object Detection | Recall=0.433 |
| Mammogram-Mass Analyzer [10] (YOLOv5L ensemble, 93M) | VinDr-Mammo | Object Detection | Recall=**0.55** |

### 7.3 Experimental Results

Table 2: My Segmentation Models Performance on BUSI Dataset

| Model & Repo | Dataset(s) | Task | Key Metrics |
|---|---|---|---|
| Medium Attention U-Net (17M) | BUSI (Ultrasound) | Segmentation | Dice=0.6084, IoU=0.5553, F1=0.6680 |
| U-Net (ResNet34 + Attention) (24M) | BUSI (Ultrasound) | Segmentation | Dice=0.7149, IoU=0.6863, F1=0.7678 |
| UNet++ (EfficientNet-B4) (20M) | BUSI (Ultrasound) | Segmentation | Dice=0.7567, IoU=0.7225, F1=0.8085 |
| MedSAM (ViT-B, 91M) | BUSI (Ultrasound) | Segmentation | Dice=0.9332, IoU=0.8772, F1=0.9332 |
| MedSAM (ViT-B + hyperparameter optimization, 91M) | BUSI (Ultrasound) | Segmentation | Dice=**0.9339**, IoU=**0.8825**, F1=**0.9362** |

Table 3: Performance comparison of classification models on CBIS-DDSM dataset

| Model & Repo | Dataset(s) | Task | Key Metrics |
|---|---|---|---|
| Med-SA Adapter (ViT-S/16 + adapters, 22M) | CBIS-DDSM (Mammography) | Classification | Accuracy=0.6296, ROC AUC=0.6692 |
| YOLOv8m-cls (15M) | CBIS-DDSM (Mammography) | Classification | Accuracy=**0.6587**, ROC AUC=**0.7262** |

## 8 Discussion and Conclusion

This comprehensive analysis reveals several key insights into the current state of deep learning for breast cancer analysis:

### 8.1 Performance Trends

The highest performing models consistently incorporate attention mechanisms, advanced backbone architectures (EfficientNet, Vision Transformers), and ensemble methods. The experimental results demonstrate that attention-based V-iT (MedSam) variants achieve superior segmentation performance (Dice: 0.93, IoU: 0.88) compared to baseline architectures. all code is available here: Project Repository

## 8.2 Reproducibility Crisis

A significant finding is the substantial gap between published results and reproducible implementations. Of the 55+ papers surveyed, only a minimal fraction provided accessible code, highlighting a critical challenge in medical AI research.

## 8.3 Clinical Applicability

While many models achieve high accuracy on benchmark datasets, their translation to clinical practice requires additional validation on diverse populations and integration with clinical workflows. The best-performing repositories in The evaluation demonstrated robust pipelines but would benefit from more comprehensive clinical validation.

## 8.4 Conclusion

Deep learning approaches show tremendous promise for improving breast cancer detection and classification. However, the field would benefit from increased focus on reproducibility, standardized evaluation protocols, and clinical validation. The experimental results contribute to this effort by providing openly available implementations and comprehensive performance benchmarks.

# 9 Future Work

- Development of multi-modal approaches combining mammography, ultrasound, and clinical data
- Investigation of federated learning to address data privacy concerns
- Enhanced explainability methods for clinical trust and adoption
- Robustness testing across diverse demographic groups and imaging devices
- Integration with clinical decision support systems
- Exploration of self-supervised and semi-supervised learning to reduce annotation requirements

# Appendix: Survey of Breast Cancer Papers (55+ Articles)

Table 4: Comprehensive List of Breast Cancer Papers

| **Paper Title** | **Reported Accuracy** | **Has Code** |
|---|---|---|
| Learning to Segment Breast Biopsy Whole Slide Images | IoU: 44 | Yes |
| Y-Net: Joint Segmentation and Classification for Diagnosis of Breast Biopsy Images | 62.5% | Yes |
| Convolution Neural Network for Breast Cancer Detection and Classification Using Deep Learning | 98.5% | No |
| Breast Cancer Multi-classification from Histopathological Images with Structured Deep Learning Model | 93.2% | No |
| Comparison of Classification Success Rates of Different Machine Learning Algorithms in the Diagnosis of Breast Cancer | 99% | No |
| Using hybrid pre-trained models for breast cancer detections | 96.3% | No |
| Early Breast Cancer Detection Based on Deep Learning: An Ensemble Approach Applied to Mammogram | 90.1% | Partial |
| Breast Cancer Diagnosis Using YOLO-Based Multiscale Parallel CNN and Flattened Threshold Swish | 98.7% | No |
| Attention-Based Deep Learning System for Classification of Breast Lesions - Multimodal, Weakly Supervised Approach | 83.6% | No |
| Self-Supervised Deep Learning to Enhance Breast Cancer Detection on Screening Mammography | 76.3% | No |
| Deep Learning Based Breast Cancer Detection Using Decision Fusion | 99% | No |
| Development of an Artificial Intelligence-Based Breast Cancer Detection Model by Combining Mammograms and Medical Health Records | 84.5% | No |
| Optimized Stacking Ensemble Learning Model for Breast Cancer Detection and Classification Using Machine Learning | 99% | No |
| Research on Logistic Regression Algorithm of Breast Cancer Diagnose Data by Machine Learning | 96.5% | No |
| Machine Learning Classification Techniques for Breast Cancer Diagnosis | 98.7% | No |
| Predicting Breast Cancer Leveraging Supervised Machine Learning Techniques | 99.12% | No |
| Machine learning-based diagnosis of breast cancer utilizing feature optimization technique | 98.77% | No |
| Breast cancer diagnosis based on feature extraction using a hybrid of K-means and support vector machine algorithms | 97.38% | No |

| **Paper Title** | **Reported Accuracy** | **Has Code** |
|---|---|---|
| Enhancing Breast Cancer Diagnosis in Mammography: Evaluation and Integration of Convolutional Neural Networks and Explainable AI | 95% | No |
| Multimodal Deep Learning for Breast Cancer Detection: Integrating Transfer Learning with Clinical Data for Enhanced Accuracy | 90.8% | No |
| YOLO Based Breast Masses Detection and Classification in Full-Field Digital Mammograms | 95.5% | No |
| Breast cancer classification based on hybrid CNN with LSTM model | 99% | No |
| An Improved Hybrid Feature Reduction for Increased Breast Cancer Diagnostic Performance | 96.31% | No |
| EfficientNet-Based Mammography Image Classification for Early Detection | 75% | No |
| Early Breast Cancer Detection Using Artificial Intelligence Techniques Based on Advanced Image Processing Tools | 99% | No |
| Breast Cancer Detection Using Machine Learning Algorithms | 94.74% | No |
| Patch-based system for Classification of Breast Histology images using deep learning | 87.4% | No |
| Automated Breast Cancer Detection Models Based on Transfer Learning | 89.5% | No |
| Deep Learning Model for Classification of Breast Cancer | 87.5% | No |
| Deep Learning RN-BCNN Model for Breast Cancer BI-RADS Classification | 85.9% | No |
| Breast Cancer Detection Using Deep Learning and IoT Technologies | 87.84% | No |
| Transfer Learning From Convolutional Neural Networks for Computer-Aided Diagnosis: A Comparison of Digital Breast Tomosynthesis and Full-Field Digital Mammography | 89% | No |
| Mass detection in digital breast tomosynthesis data using convolutional neural networks and multiple instance learning | 89.81% | No |
| Intelligent Hybrid Deep Learning Model for Breast Cancer Detection | 86.21% | No |
| A deep learning approach for the analysis of masses in mammograms with minimal user intervention | 90% | No |
| Discrimination of Breast Cancer with Microcalcifications on Mammography by Deep Learning | 87.3% | No |
| SD-CNN: A shallow-deep CNN for improved breast cancer diagnosis | 90% | No |
| Improving Breast Cancer Detection using Symmetry Information with Deep Learning | AUC: 0.92 | No |

| **Paper Title** | **Reported Accuracy** | **Has Code** |
|---|---|---|
| Large scale deep learning for computer aided detection of mammographic lesions | Not specified | No |
| Detection of masses in mammograms using a one-stage object detector based on a deep convolutional neural network | Not specified | No |
| Applying Explainable Machine Learning Models for Detection of Breast Cancer Lymph Node Metastasis in Patients Eligible for Neoadjuvant Treatment | AUC: 0.78 | No |
| Thermal-based early breast cancer detection using inception V3, inception V4 and modified inception MV4 | 99% | No |

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